

Active Directory Project w/ Azure

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Setting up a Domain Controller VM

Important Note: All images for this project can be found in the “Active_Directory_Project” Folder. By sorting files by date created, in ascending order; all sections and screenshots can be seen chronologically, making the project easier to follow. Additionally, these screenshots show the location of objects that were interacted with in each step using the color green. Other colors are also used like yellow, which signifies that something is optional, and red which signifies that an error/ unexpected outcome has occurred. If attempting to do this project, it’s recommended to use the “Active_Directory_Project” images if you get stuck or want to double check your progress.

1. Create a Virtual Machine (VM)

- Log into the Azure portal
- Select “**All Services**” in the left pane, then Select “**Virtual Machines**”
- Select “**Create**”, “**Presets**”, and then “**Continue to create a VM**”

2. Basics Tab (Important)

- **Subscription:** Leave default
- **Resource Group:** Select “**Create new**” and name (e.g. ad-lab-rg)
- **Virtual Machine Name:** DC01
- **Region:** Select Closest location to you
- **Availability Options:** Leave default

- **Image:** Windows server 2025 Datacenter -x64 Gen2 (free service eligible)
- **Size:** 2Bs or similar

3. Administrator Account

- **Username:** Any
- **Password:** Strong Password (save this)

* This local admin account will become the domain administrator later*

4. Inbound Port Rules

- **Public Inbound Ports:** Select “Allow selected ports”
- **Select Inbound Ports:** RDP (3389)

5. Disk Tab

- **OS Disk Type:** Select “Standard SSD (locally-redundant storage)”
- All other setting default

6. Networking Tab

- **Virtual Network:** Create new (e.g. DC01-Clients-vnet)
- **Public IP:** Create new (e.g. AD01-ip)
- **NIC network security group:** Basic

7. Review + Create

- Review settings
- Select “Create”
- Wait for deployment to complete

Assigning a Static IP

WHY: Assigning a static IP prevents IP changes that can break DNS or Active Directory later.

1. Accessing DC01 Setting

- Select “**All Services**” in the left pane, then Select “**Virtual Machines**”
- Select **DC01**, highlighted in blue, under the “Virtual machines” tab

2. Accessing DC01’s Network Interface

- In the DC01 pane, open “**Networking**” and select “**Network Settings**”
- Locate “**Network interface: dc01955_z1**”, Select “**dc01955_z1**”

3. Assign IP to Static

- Under the dc01955_z1 pane, open “**Settings**”, and select “**IP configurations**”
- Select “**ipconfig1**”
- A new side tab will appear. Under “**Private IP address settings**”, Select “**Static**”
- Lastly, **Save**

Installing Active Directory Domain services

Part 1: Connecting to VMs With RDP

1. Accessing DC01 Setting

- Select “**All Services**” in the left pane, then Select “**Virtual Machines**”
- Select **DC01**, highlighted in blue, under the “Virtual machines” tab

2. Confirm Start-up

- Check that the “Start” button is dull and not blue
- If “Start” is blue, Select it to initiate VM deployment

3. Connecting to DC01 Through RDP

- In DC01, under the “Properties”, locate “Public IP address”
- **Copy** the IP address (e.g. 40.81.16.140)
- In Window’s searchbar, open “**Run**”
- In the run application, type “**mstsc**” and select “**OK**”
- The “Remote Desktop Connection” page will open. **Paste** the IP in for “**Computer**” and enter user (created with the VM) in “**User name**”
- **Optional:** Select “**Save As**” to create an RDP shortcut for future use
- Select “**Connect**”. Do this twice
- Enter Admin Username and Password created with VM and Select “**OK**”
- A warning will pop up. We don’t care! Select “**Yes**”
- Connection to DC01 is complete

Part 2: Domain Services Setup

1. Open Server Manager

- Server Manager should pop-up automatically after login
- If not, open it from **Start Menu**

2. Add Roles and Features

- Select “**Add roles and features**”
- On the “**Before you begin**” screen, Select “**Next**”
- Select “**Role-based or feature- based Installation**” and “**Next**”
- Select the local server (default) and select “**Next**”

3. Server Roles

- Check “**Active Directory Domain Services**”
- When prompted, select “**Add Features**”, then select “**Next**”

4. Features

- Leave Default, Select “**Next**”

5. AD DS Information

- Review and Select “**Next**”

6. Confirmation

- Confirm the role selection
- Select “**Install**”

*Wait for the install to complete. **Don't Reboot** yet and **Do Not Close** the “**Results**” tab*

Promoting Server to Domain Controller

1. Start Promotion

- From the previous section on the “**Results**” tab, Select “**Promote this server to a domain controller**”
- Alternatively, if the “**Results**” tab is close, Open “**Server Manager**”, select the **notification flag** at the top of the page, then select “**Promote this server to a domain controller**”

2. Deployment Configuration

- Select “**Add a new forest**”
- Enter a “**Root domain name**” (e.g. lab.local) and select “**Next**”

This lab uses a private internal domain. This is fine for a lab! Just note that in real environments, domain naming requires more planning

3. Domain Controller Options

- **Forest functional level:** Leave default
- **Domain function level:** Leave default
- **DNS server:** Check
- **Global Catalog:** Check
- Set the **DSRM** password and **save** it. Used for active directory recovery scenarios
(Hopefully Rare!)

4. DNS Options

- Warning is expected in lab environments. Ignore it and select “**Next**”

5. Addition Options

- Accept NetBIOS domain name and select “**Next**”

6. Paths

- Leave default and select “**Next**”

7. Review Options

- Review and select “**Next**”

8. Domain Name System (DNS) server

- Wait for check to complete
- Warning are okay in lab environment
- Select “**Install**”

9. Reboot and log in

- After install, reboot will automatically take place
- Sign back into the **DC01** VM with **domain administrator account**

Creating Organizational Units

1. Open Active Directory Users and Computers

- From Server Manager, Select “Tools” on the top right, then Select “Active Directory Users and Computers”
- OR search and Select “Active Directory Users and Computers” from the start menu

2. Locate Domain Root

- In the left pane, expand the domain (e.g lab.local)
- This top level-container is the domain root

It’s important to create a top-level OU to represent an organizational structure. Don’t place objects directly at the root.

3. Create the **_Branches** OU

- Right- click the domain root (e.g. lab.local)
- Select “**New**”, followed by “**Organizational Units**”
- Name the new OU “**_Branches**”
- Ensure “**Protect container from accidental deletion**” is checked
- Select “**Ok**”

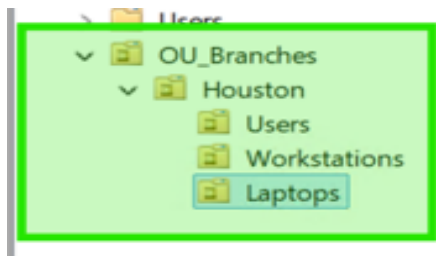
4. Create Branch OU

Now, an OU is created for a specific Branch. In a real business application, there would be a branch for each physical location that a company owns.

- Right-click “_Branches” OU
- Select “New”, followed by “Organizational Units”
- Name the new OU “Houston”
- Right-click “Houston” OU
- Select “New”, followed by “Organizational Units”
- Name the new OU “Users”
- Repeat these steps to create “Workstations” and “Laptops” OUs under “Houston”

5. Verify OU Structure

- Confirm OUs exist and are properly nested
- What the structure should look like:



6. Key Notes

- Avoid placing users or computers in default containers
- Group policy is applied at the OU level (never applied on containers)
- This structure is very important for future steps!

Creating Users

1. Locate Users OU in Active Director Users and Computers

- Log into Domain Controller
- Open “**Active Director Users and Computers**”
- Expand domain (e.g. lab.local)
- Navigate to **_Branches, Houston,** then **Users**

2. Create The First User

- Right-Click “**Users**” OU
- Select “**New**”, then “**Users**”
- **First Name:** Alice
- **Last Name:** Johnson
- **User Login Name:** ajohnson
- Select “**Next**”

3. Set User Password

- Set initial Password
- Uncheck “**Users must change password on next login**” (For lab only)
- Uncheck “**Users cannot change password**”
- Uncheck “**Password never expires**”

4. Create Additional Users

Repeat steps 3-4 to create two more accounts the **/Houston/Users** OU

- Bob Martinez -bmartinez
- Chris Walker- Walker

5. Verify User Creating

- Confirm all users appear in /_Branches/Houston/Users
- Double-click a user to view account properties
- Verify that the Distinguished Name reflects the correct OU Path

6. PowerShell Alternative For Creating Users (Optional)

- Open “Notepad” from “Startup Menu”
- Enter the following script:

```
$ou = "OU=Users,OU=Houston,OU=_Branches,DC=lab,DC=local"
```

```
New-ADUser -Name "Alice Johnson" -GivenName Alice -Surname Johnson -SamAccountName ajohnson  
-Path $ou -AccountPassword (Read-Host -AsSecureString) -Enabled $true
```

```
New-ADUser -Name "Bob Martinez" -GivenName Bob -Surname Martinez -SamAccountName bmartinez  
-Path $ou -AccountPassword (Read-Host -AsSecureString) -Enabled $true
```

```
New-ADUser -Name "Chris Walker" -GivenName Chris -Surname Walker -SamAccountName cwalker  
-Path $ou -AccountPassword (Read-Host -AsSecureString) -Enabled $true
```

- Select “File” then “Save As”
- **File Name:** users.bat OR users.ps1
- From “Startup Menu”, Open “PowerShell as Administrator”
- At the prompt, enter:

Line 1: cd C:\Users\(Account Name)\Documents

Line 2: .\users.bat

PowerShell is common for automation and bulk user provisioning

Creating Security Groups

1. Create a Centralized _Groups OU

- Open “**Active Director Users and Computers**”
- Right- click the domain root (e.g. lab.local)
- Select “**New**”, followed by “**Organizational Units**”
- Name th OU “**_Groups**”
- Ensure “**Protect container from accidental deletion**” is checked
- Select “**Ok**”

2. Create the Helpdesk Group

- Right-click “**_Groups**” OU
- Select “**New**”, followed by “**Group**”
- **Group Name:** Helpdesk
- **Group Scope:** Global
- **Group Type:** Security
- Select “**OK**”

3. Create Remaining Security Groups

Repeat step 2 to create the following Security Groups:

- Accounting
- ITSupport

4. Verify Group Creation

- Confirm all groups exist in the “**_Groups**” OU

- Double-click a user to view account properties
- Verify that the Distinguished Name reflects the correct OU Path

5. Add Users to Group

Assign users to groups based on their roles:

Add Alice Johnson to Helpdesk

- Open the “**Helpdesk**” Group
- Select the “**Members**” tab
- Select “**Add**”
- Add user (ajohnson) and select “**Check Name**”
- Select “**Ok**”
- Select “**Apply**”, then “**Ok**” again

Add remaining users

- Add bmartinez to Accounting
- Add cwalker to ITSupport

6. Verify Group Membership

- Open each group and confirm correct membership

7. Using PowerShell to Create Security Groups & Membership (Optional)

- Open “**Notepad**” from “**Startup Menu**”
- Enter the following script:

```
$groupsOU = "OU=_Groups,DC=lab,DC=local"
```

```
New-ADGroup -Name "Helpdesk" -GroupScope Global -GroupCategory Security -Path $groupsOU
```

```
New-ADGroup -Name "Accounting" -GroupScope Global -GroupCategory Security -Path $groupsOU
```

```
New-ADGroup -Name "ITSupport" -GroupScope Global -GroupCategory Security -Path $groupsOU
```

```
Add-ADGroupMember -Identity "Helpdesk" -Members ajohnson
```

```
Add-ADGroupMember -Identity "Accounting" -Members bmartinez
```

```
Add-ADGroupMember -Identity "ITSupport" -Members cwalker
```

- Select **“File”** then **“Save As”**
- **File Name:** security_groups.bat
- From **“Startup Menu”**, Open **“PowerShell as Administrator”**
- At the prompt, enter:

Line 1: cd C:\Users\(Account Name)\Documents

Line 2: .\security_groups.bat

8. Key Notes

- Users gain access through group membership
- Groups are generally centralized, not branch-based
- The structure supports scalability and delegation

Joining a ClientVM to the Domain

Part 1: Create Client Virtual Machine (VM)

1. Initiation

- Log into the Azure portal
- Select “**All Services**” in the left pane, then Select “**Virtual Machines**”
- Select “**Create**”, “**Presets**”, and then “**Continue to create a VM**”

2. Basics Tab (Important)

- **Subscription:** Leave default
- **Resource Group:** Select “**Resource Group**” used for “**DC01**” (e.g. ad-lab-rg)
- **Virtual Machine Name:** CLIENT01
- **Region:** Select Closest location to you
- **Availability Options:** Leave default
- **Image:** Windows server 2025 Datacenter -x64 Gen2 (free service eligible)
- **Size:** 2Bs or similar

3. Administrator Account

- **Username:** Same as username created for “**DC01**”
- **Password:** Same as password created for “**DC01**”

4. Inbound Port Rules

- **Public Inbound Ports:** Select “Allow selected ports”
- **Select Inbound Ports:** RDP (3389)

5. Networking Tab

- **Virtual Network:** Must be the same as “DC01” (e.g. DC01-Clients-vnet)
- **Public IP:** Create new (e.g. CLIENT-ip)
- **NIC network security group:** Basic

6. Review + Create

- Review settings
- Select “Create”
- Wait for deployment to complete

Part 2: Connecting to CLIENT01 With RDP

1. Accessing CLIENT01 Setting

- Select “All Services” in the left pane, then Select “Virtual Machines”
- Select **CLIENT01**, highlighted in blue, under the “Virtual machines” tab

2. Confirm Start-up

- Check that the “Start” button is dull and not blue
- If “Start” is blue, Select it to initiate VM deployment

3. Connecting to CLIENT01 Through RDP

- In CLIENT01, under the “Properties”, locate “Public IP address”
- **Copy** the IP address (e.g.55.229.240.184)
- In Window’s searchbar, open “Run”
- In the run application, type “mstsc” and select “OK”

- The “Remote Desktop Connection” page will open. **Paste** the IP in for “**Computer**” and enter user (created with the VM) in “**User name**”
- **Optional:** Select “**Save As**” to create an RDP shortcut for future use
- Select “**Connect**”. Do this twice
- Enter Admin Username and Password created with VM and Select “**Ok**”
- A warning will pop up. We don’t care! Select “**Yes**”
- Connection to CLIENT01 is complete

Part 3: Configure CLIENT01

1. Configure DNS

- From Start-up Menu, open “**Network Connections**”
- Right-click “**Ethernet Network**”, select “**Properties**”
- Select “Internet Protocol Version 4 (TCP/IPv4)”
- Select “**Properties**”
- Back in Azure, copy the private IP of DC01 (e.g. 10.0.0.4)
- Return to CLIENT01, check “Use the following DNS server address”
- Paste the private IP of DC01 into “**Preferred DNS server:**”
- Select “**Ok**”
- Select “**Close**”

2. Join the Domain

- Open “**Settings**”, select “**System**”, then “**About**”
- Located by “**Related Links**”, select “**Domain or workgroup**”
- In the “**Computer Name**” tab, select “**Change**”

- **Computer Name:** CLIENT
- **Member of:** check “**Domain**”, input “**lab.local**”
- Select “**Ok**”
- Enter admin credentials, select “**Ok**”
- When prompted with “**Welcome to the lab.local domain**”, select “**Ok**”
- Select “**Ok**” one last time

3. Verify Domain Join

- After Reboot, log into “**CLIENT01**” as a domain user account through RDP
- Open “**CMD**” as Administrator
- Type the following command: “**dsregcmd /status**”
- Under “**Device State**”, Confirm device shows as domain joined
- Locate “**CLIENT**” in “**Active Directory Users and Computers**”

*By default, computers joined to a domain appear in the “**Computer**” container.*

Part 4: What Happens During a Domain Join (Informational)

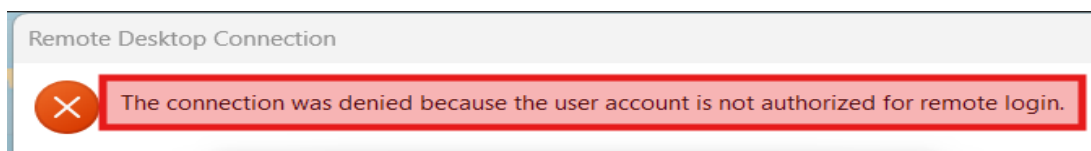
1. A computer account is created in Active Directory
2. A secure trust relationship is established with the domain
3. The client begins authenticating against the domain controller
4. Group policy becomes applicable to computers that join the domain

Login Test

1. Attempting Sign-In as “ajohnson”

- From **Azure** homepage, Select “**All Services**” in the left pane, then Select “**Virtual Machines**”
- Select **CLIENT01**, highlighted in blue, under the “Virtual machines” tab
- In **CLIENT01**, under the “Properties”, locate “Public IP address”
- **Copy** the IP address (e.g.55.229.240.184)
- In Window’s searchbar, open “**Run**”
- In the run application, type “**mstsc**” and select “**OK**”
- The “Remote Desktop Connection” page will open. **Paste** the IP in for “**Computer**” and enter user (created with the VM) in “**User name**”
- Select “**Connect**”. Do this twice
- **This is where things change!** Select “**Other user**”
- Enter “**LAB\ajohnson**” for the username, then enter **ajohnson’s Password** and Select “**Ok**”
- Select “**YES**”

Error Message:



2. Trouble Shooting RDP Connection Domain Users

- Using **RDP**, log into “**CLIENT01**” through the administrator account

First, ensure that the NetBIOS name is “LAB”:

- Open “**Command Line**” as administrator.
- In the Command Prompt, type “**nbtstat -n**”
- Under “**NetBIOS Local Name Table**”, verify that:
 - “**LAB**” is under the “**Name**” field
 - “**LAB**” is Registered under “**Status**” field

Check If Specific Groups/Users Have RPD Permissions

- Open “**Settings**”, select “**System**”, scroll down and select “**About**”, then select “**Remote Desktop**”
- Select “**Remote Desktop users**”. A new page will appear. Select “**Add**”
- Input “**Helpdesk**” under “**Enter the object names to select:**”
- Select “**Check Names**”
- Enter **administrator username** and **password** when prompted
- Select “**Ok**”
- Repeat this process to add “**Accounting**” and “**ITSupport**”
- Select “**Ok**”
- Exit the “**CLIENT01**” VM

3. Verify Successful Login

- Reconnect via RDP to Client01 using “**LAB\ajohnson**”, “**LAB\bmartinez**”, or “**LAB\cwalker**”
- For specific steps to establish RDP session, return to step 1

- On Client01, open “**Command Line**” as administrator, and enter to following command: **whoami**
- Review the command output and verify that the user listed matches who signed into “**Client01**”

Bonus Tasks

Part 1: Configuring Password Policies

1. Navigating to Password Policy

- Login into **DC01** with admin credential through RDP
- In **start-up menu**, search and open “**Group Policy Management**”
- Expand “**Forest: lab.local**”, “**Domains**”, then “**lab.local**”
- Right-click “**Default Domain**” and select “**Edit**”
- Expand “**Computer Configuration**”, “**Policies**”, “**Windows Settings**”, “**Security Settings**”, then “**Account Policy**”
- Select “**Password Policy**”

2. Configure Password Policy

- **Minimum Password Length**: 13 characters
- **Apply** and select “**Ok**”
- **Maximum Password Age**: Set to 180 days or uncheck “**Define this policy setting**”
- **Apply** and select “**Ok**”
- **Password must meet complexity requirements**: Disabled
- **Apply** and select “**Ok**”

3. Policy Justification

This password policy balances security, usability, and administrative efficiency. A 13-character minimum password length provides strong protection against

password-cracking attacks, while a 180-day password age avoids the negative consequences of frequent password changes (Password reuse, predictable password patterns, increased helpdesk cost, and unsafe storage practices). Disabling complexity requirements encourages the use of long passphrases that are easier for users to remember and harder for attackers to compromise.

Part 2: Creating a Logon Script

1. Writing Logon Script

- Login into **DC01** with admin credential through RDP
- In **start-up menu**, search and open “**Notepad**”
- **Type:** echo Welcome to the Domain!

2. Convert to .bat and Saving Script

- Select “**File**”, then “**Save as**”
- Rename file as “**logon.bat**”
- Edit file path to: “**C:\Windows\SYSVOL\sysvol\lab.local\scripts**”
- Open “Active Directory Users and Computers”
- Locate to “**Users**” under “**Houston**” branch
- Select “**Alice Johnson**”
- Select “**Profile**” tab,
- **Login Script:** logon.bat
- **Apply** and select “**Ok**”

Part 3: Delegating Password Rest to Helpdesk

1. Opening Delegate Control on “Users”

- Login into **DC01** with admin credential through RDP
- In **start-up menu**, search and open “**Active Directory Users and Computers**”
- Right-click “**Users**” under the “**Houston**” Branch and select “**Delegate Control**”

2. Configuring Delegation Control Through Wizard

- The Wizard page will automatically pop up
- On the welcome screen, select “**Next**”
- Select “**Add**”
- Type “**Helpdesk**” under “**Select the object names to be selected**”
- Select “**Check Names**”, then select “**Ok**”

This is one of the most common real-world delegation scenarios for helpdesk teams

Part 4: Moving Client VM to the Correct Branch OU

1. Moving Client01 to Workstations

- (if not signed in) Login into **DC01** with admin credential through RDP
- In **start-up menu**, search and open “**Active Directory Users and Computers**”
- Expand “**lab.local**”, “**_Branches**”, and “**Houston**” to reveal “**Workstations**”
- Select “**Computers**” under “**lab.local**” to reveal “**CLIENT01**”
- Drag “**CLIENT01**” to “**Workstations**”
- Prompt will appear. Select “**Yes**”
- Verify “**CLIENT01**” is Located in “**Workstations**” by clicking “**Workstations**”